NUMERICAL METHODS IN COSMOLOGY

ELLIOT MARSHALL

1. Introduction

The purpose of these notes is to give a brief introduction to numerically solving the Einstein equations in a cosmological context. An accompanying jupyter notebook implementing some of the methods discussed here is available here. Note this binder link can take a few minutes to start up!

If you find any typos/errors let me know at `elliot.marshall@iacm.forth.gr`.

2. Finite Difference Methods

Although derivative operators act on continuum functions, there are many situations where we will only have at data at finitely many discrete points. For example, if we have taken measurements in an experiment or are we are solving differential equations on a computer, we only have access to a discrete approximation of a continuum function. However, in these situations we still may want to find (an approximation of) the derivative. One way to do this is using *finite difference* approximations.

Remark 2.1 (Useful Resources). There are numerous textbooks on the finite difference method. I particularly like the following:

- *Finite Difference Methods for Ordinary and Partial Differential Equations* by Randall J. Leveque [51],
- *Time-Dependent Problems and Difference Methods* by Bertil Gustaffson, Heinz-Otto Kreiss, and Joseph Oliger [42],
- *Initial-Boundary Value Problems and the Navier-Stokes Equations* by Heinz-Otto Kreiss and Jens Lorenz [47].

The first two books are more focused on the numerical analysis aspect of finite difference methods. The third book is focused on (continuum) PDE analysis but contains some nice proofs on the existence of solutions using finite difference approximations, for example in Chapter 3.

2.1. Discrete Approximations of Derivatives.

Finite difference operators are the simplest method for approximating derivatives and solving differential equations numerically. The fundamental idea is that we can replace continuum derivative operators with discrete algebraic expressions obtained from truncated Taylor series expansions. As an example, let us derive the basic finite difference approximations for the first derivative.

Suppose we have a (suitably) smooth scalar function $u(x)$,

$$u : [a, b] \to \mathbb{R}, \quad a, b \in \mathbb{R}, \quad a < b.$$

The Taylor series approximation of $u(x + \Delta x)$, $\Delta x > 0$, about the point x is given by

$$u(x + \Delta x) = u(x) + (\Delta x)u^{(1)}(x) + \frac{(\Delta x)^2}{2}u^{(2)}(x) + \frac{(\Delta x)^3}{6}u^{(3)}(x) + \cdots \tag{2.1}$$ eqn:Taylor_Series_F

where $u^{(n)}$ denotes the nth derivative. Now, by re-arranging (2.1) for $u^{(1)}(x)$ we obtain the expression

$$u^{(1)}(x) = \frac{u(x + \Delta x) - u(x)}{\Delta x} + \mathcal{O}(\Delta x).$$

Hence the expression

$$\frac{u(x+\Delta x)-u(x)}{\Delta x} \tag{2.2}$$ eqn:Forward_Differe

gives us a first order approximation (in Δx) for the first derivative using only the values of the function at the points x and $x+\Delta x$. By first order, we mean the error in our approximation will decrease linearly as $\Delta x \to 0$. On an intuitive level, this approximation makes sense; by taking the limit $\Delta x \to 0$, we recover the definition of the derivative at the point x.

Hence, the operator (2.2) is known as a first order *forward difference*. Similarly, we can take the Taylor expansion of $u(x-\Delta x)$,

$$u(x-\Delta x) = u(x) - (\Delta x)u^{(1)}(x) + \frac{(\Delta x)^2}{2}u^{(2)}(x) - \frac{(\Delta x)^3}{6}u^{(3)}(x) + \cdots \tag{2.3}$$ eqn:Taylor_Series_E

to obtain the first order *backward difference*

$$\frac{u(x)-u(x-\Delta x)}{\Delta x} \tag{2.4}$$ eqn:Backward_Differ

In practice, we would like approximations whose error decreases faster than $\mathcal{O}(\Delta x)$[1]. In fact, we can derive such approximations in a similar manner to the backwards and forwards differences.

Subtracting (2.3) from (2.1) yields

$$u(x+\Delta x) - u(x-\Delta x) = 2(\Delta x)u^{(1)}(x) + \frac{(\Delta x)^3}{3}u^{(3)}(x) + \cdots$$

Solving for $u^{(1)}(x)$ we then obtain the expression

$$u^{(1)}(x) = \frac{u(x+\Delta x)-u(x-\Delta x)}{2\Delta x} + \mathcal{O}\big((\Delta x)^2\big)$$

Hence, we have obtained a second order accurate finite difference approximation of the first derivative $u^{(1)}(x)$ using the values at the points $x+\Delta x$ and $x-\Delta x$. The expression

$$\frac{u(x+\Delta x)-u(x-\Delta x)}{2\Delta x} \tag{2.5}$$ eqn:central_differe

is known as a second order *central difference* stencil. By incorporating the Taylor series expansions of additional points, e.g. $u(x+2\Delta x)$ and $u(x-2\Delta x)$, one can obtain a higher order approximation of the first derivative. Approximations for higher derivatives can also be derived using similar calculations.

Remark 2.2 (Regularity Assumptions). It should be noted that our derivation of the difference stencils (2.2), (2.4), and (2.5) relied on the function u being suitably differentiable. However in certain situations, for example when shocks form, u may no longer be differentiable. In this case, one can use *finite volume* methods to handle discontinuities in the solution.

Now, we can discretise our domain $[a,b]$ into a uniform grid $\{x_i\}$ of $N+1$ points

$$x_i = x_0 + i(\Delta x), \quad i = 0, 1, \cdots, N, \tag{2.6}$$ {?}

where $x_0 = a$, $x_N = b$, and $\Delta x := x_i - x_{i-1}$ is the *spatial step size*. See Figure 1 for a diagram.

FIGURE 1. A discretisation of the domain $[a,b]$ into a uniform grid with $N+1$ points. Each grid point is Δx apart.

[1]Typically (formally) second order accurate approximations are the minimum used in practice.

We can use the finite difference stencil (2.5) to approximate the first derivative at a point x_i, using the values of u at adjacent grid points,

$$u_i^{(1)} = \frac{u_{i+1} - u_i}{2\Delta x}, \tag{2.7}$$ eqn:central_differe

where $u_i = u(x_i)$. Thus, we have a finite difference approximation for the derivative at a grid point x_i. Based on our Taylor series derivation, we expect that the error in the approximation (2.7) should decrease like $(\Delta x)^2$ as the grid size $\Delta x \searrow 0$.

Following this example, a finite difference scheme for a general partial differential equation (PDE) can then be built by discretising in *both* time and space, see Figure 2. Derivatives are then replaced with their corresponding finite difference approximations. This results in a system of algebraic equations which we can implement numerically.

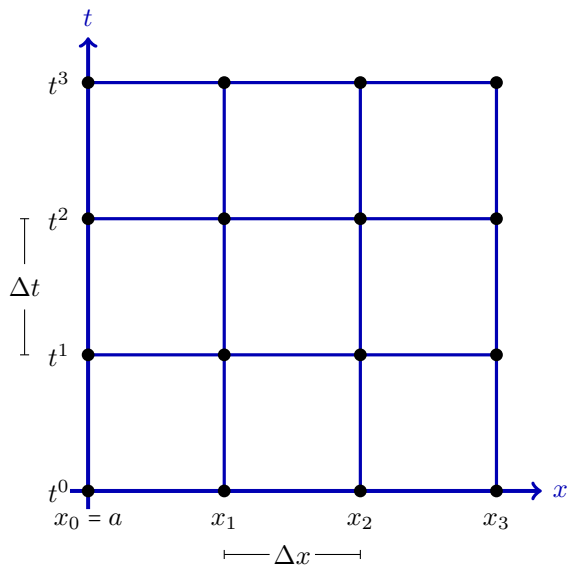


FIGURE 2. A uniform discretisation of the space-time domain with timestep Δt and spatial step Δx.

Example 2.3 (Upwind Method for the Advection Equation). Consider the 1+1 advection equation,

$$\partial_t u + \partial_x u = 0.$$

We can discretise both the time and space derivatives to obtain the algebraic system

$$\frac{u^{n+1} - u^n}{\Delta t} + \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0,$$

where $u_i^n = u(t^n, x_i)$. Here, we have used a forward difference for the time derivative and a backwards difference for the spatial derivative. Re-arranging for u_i^{n+1} we obtain the first order accurate update formula,

$$u_i^{n+1} = u_i^n + \frac{\Delta t}{\Delta x}(u_i^n - u_{i-1}^n). \tag{2.8}$$ eqn:upwind_advectio

The scheme (2.8) is commonly known as the *upwind* method. Thus, given the values of $u(x_i)$ and $u(x_{i-1})$ at time t^n, we can the value of $u(x_i)$ at time t^{n+1}.

It turns out that to obtain a stable[2] numerical scheme, we need to enforce a condition on the size of the timestep,

$$\Delta t = \lambda \Delta x, \quad 0 < \lambda < 1.$$

This ensures that the computational domain of dependence is contained in the analytic domain of dependence. Such conditions were first derived in the work of Courant, Friedrichs, and Lewy and are commonly referred to as 'CFL' restrictions in the literature.

Remark 2.4 (Discrete Boundary Conditions). Note that in order to apply the backward stencil (2.4) at the boundary point x_0 in (2.8), we would require the value of u at the grid point x_{-1} which is not included in the spatial domain. There are two standard ways to handle this problem. The first is to employ one-sided stencils near the boundary which only use the values of the function from the interior of the domain. The second option is to use the external boundary conditions to fill in the values of 'ghost' points, such as u_{-1}, which lie outside the spatial domain.

2.1.1. *Exercises.*

(1) Show that $\frac{u_{i+1}-2u_i+u_{i-1}}{(\Delta x)^2}$ is a second order approximation for the second derivative.
(2) Implement the first order upwind method for the advection equation with advection speed $a > 0$ on a computer. How does the CFL restriction change for larger values of a? Why?
(3) Implement the advection equation using a central difference for the spatial derivatives instead of a backwards difference. What do you notice?

A basic implementation of the upwind method is given in the Jupyter notebook linked in the introduction. Exercises 2 and 3 can be done by minor modifications to the existing python code.

2.2. **Convergence, Consistency, and Stability.** In the vein of [26], let us consider a $1+1$ PDE of the form

$$Lu = f, \quad x \in [a, b], \tag{2.9}$$ eqn:P2_Background_F

where L is a differential operator, u is the function (or possibly vector of functions) we are trying to solve for, and f is a source term. For simplicity, we will only consider problems with periodic boundary conditions

$$u(t, a) = u(t, b). \tag{2.10}$$ {?}

To solve the system (2.9) numerically, we introduce a discrete uniform grid in time and space,

$$x_j = x_0 + jh, \quad j = 0, 1, ..., N, \tag{2.11}$$ {?}

$$t_n = t_0 + nk, \quad n = 0, 1, ..., \tag{2.12}$$ {?}

where $h = \Delta x$ and $k = \Delta t$ are the spatial and time step sizes, respectively. For simplicity, we further assume that the ratio between spatial and time steps is constant,

$$\frac{k}{h} = \lambda \tag{2.13}$$ {?}

where λ is some fixed, positive constant. A discrete approximation of (2.9) is given by[3]

$$L^h u^h = f^h \tag{2.14}$$?eqn:P2_BG_Discrete

where u^h and f^h are discretised grid functions and L^h is a discrete differential operator.

Ultimately, we want to understand how well the discrete function u^h approximates the continuum solution u. We define the **global error** e^h by

$$e^h := u - u^h \tag{2.15}$$ eqn:GlobalError_Def

[2]Further details on stability below in the following section!
[3]Since our time step k is proportional to h, we index our discrete quantities using h alone.

where the continuum solution u is understood to be evaluated at the grid points. We say a numerical method is **convergent** if, in some appropriate discrete norm[4],

$$\|e^h\|_h \to 0 \quad \text{as} \quad h \to 0. \tag{2.16}$$ {?}

That is u^h converges to the continuum solution. Clearly, any useful numerical scheme *must* be convergent. The **truncation error** τ^h is defined by applying our discrete scheme to the continuum solution

$$\tau^h = L^h u - f^h. \tag{2.17}$$ eqn:TruncationError

In essence, this measures the error introduced by the discrete operator L^h. Our discrete scheme is said to be nth-order accurate if τ^h is $\mathcal{O}(h^n)$. A method is **consistent** if

$$\|\tau^h\|_h \to 0 \quad \text{as} \quad h \to 0. \tag{2.18}$$ {?}

While consistency is clearly necessary for convergence, it is not sufficient[5].

We call a method **stable** [69] if there are positive constants h_0, C, and α, independent of h, such that for all $t_0 < t_n$ and $h < h_0$

$$\|u^h(t_n)\|_h \le Ce^{\alpha t_n}\|u^h(t_0)\|_h, \tag{2.19}$$ {?}

which amounts to a discrete analogue of well-posedness. Moreover, for a consistent finite difference approximation of a linear PDE, the famous Lax equivalence theorem states stability and convergence are equivalent [42]. While this result no longer holds, in general, for non-linear PDE, it provides the useful intuition that stability and convergence are closely intertwined.

c:Error_Analysis_FD⟩? 2.3. **Error Analysis.** In most cases, we will not have a rigorous proof that our numerical scheme is convergent. However, in order to have confidence in our numerical results, it is vital that we can demonstrate that our scheme is converging at the expected rate. Unfortunately, since we do not, in general, know the exact continuum solution u, we cannot estimate the rate of convergence directly by computing the global error using (2.15). To get around this, we assume that the global error admits a power series expansion, known as a *Richardson expansion*, of the form

$$e^h = u - u^h = he_1 + h^2e_2 + h^3e_3 + \cdots \tag{2.20}$$ eqn:Richardson_Expa

where the e_i are continuum functions of (t, x) which are independent of the mesh size h.

Re-arranging (2.20) for u then yields

$$u = u^h + he_1 + h^2e_2 + h^3e_3 + \cdots = u^h + e^h. \tag{2.21}$$ eqn:Richardson_Expa

Assuming L^h is linear[6], by substituting (2.21) into (2.17) and using the fact that u^h is a solution of the discrete system, we then find,

$$\tau^h = L^h e^h = h^n L^h(e_n) + \mathcal{O}(h^{n+1}), \tag{2.22}$$ {?}

where we have assumed $e^h = \mathcal{O}(h^n)$ to obtain the second equality. Thus, when the global error admits a Richardson expansion, the truncation error and global error have the same leading order behaviour in h.

Now, using the Richardson expansion, we can estimate the rate of convergence by considering a sequence of discrete solutions. To begin, suppose we have numerical solutions computed at three

[4]See [49, 10.2] for a discussion on which norms are appropriate.
[5]See for example [70, 1.4].
[6]A similar relation can be shown to hold in the non-linear case, see [49].

resolutions h, $2h$, and $4h$ using a kth-order accurate scheme[7]. Taking the difference of u^{4h} and u^h, we find

$$\begin{aligned} u^{4h} - u^h &= \Big(u - \sum_{n=1}^{\infty} (4h)^n e_n\Big) - \Big(u - \sum_{n=1}^{\infty} h^n e_n\Big) \\ &= \sum_{n=1}^{\infty} (1 - 2^{2n}) e_n h^n \\ &= (1 - 2^{2k}) e_k h^k + \mathcal{O}(h^{k+1}). \end{aligned} \tag{2.23}$$ eqn:u4h_diff

Similarly, for u^{2h} we obtain

$$u^{2h} - u^h = (1 - 2^k) e_k h^k + \mathcal{O}(h^{k+1}). \tag{2.24}$$ eqn:u2h_diff

Following [62], we define the error ratio

$$R(4h, 2h; h) = \frac{u^{4h} - u^h}{u^{2h} - u^h}. \tag{2.25}$$ eqn:error_ratio_def

Now, by substituting (2.23) and (2.24) into (2.25), we find that the error ratio is given by

$$R(4h, 2h; h) = 2^k + 1$$

to leading order. The (pointwise) order of convergence is then computed[8] by

$$\log_2 \big(|R(4h, 2h; h) - 1|\big) = k. \tag{2.26}$$ eqn:Log2_error_rati

It is important to note that this analysis assumes that our reference solution u^h has a coarse resolution [51, §A.6.3]. Typically, however, we use a high resolution run as our reference. In this case, we expect that u^h should be a highly accurate approximation of the continuum solution u. Hence, for a suitably high resolution reference solution, the difference between approximate solutions can be well-approximated by

$$\begin{aligned} u^{4h} - u^h &\approx u^{4h} - u = 2^{2k} h^k e_k + \mathcal{O}(h^{k+1}), \\ u^{2h} - u^h &\approx u^{2h} - u = 2^k h^k e_k + \mathcal{O}(h^{k+1}). \end{aligned}$$

The error ratio, to leading order, is then

$$R(4h, 2h; h) = \frac{u^{4h} - u^h}{u^{2h} - u^h} = 2^k, \tag{2.27}$$ {?}

and the order of numerical accuracy is calculated by

$$\log_2 \big(|R(4h, 2h; h)|\big) = k. \tag{2.28}$$ eqn:Log2_error_rati

Observe that if we used the formula (2.26) in the high resolution situation, we would *underestimate* the order of convergence. Thus, by computing (2.28) for a sequence of resolutions, we can provide convincing evidence that our numerical scheme is converging even if we cannot directly prove it. This process is commonly known as *convergence testing*.

It is essential to emphasise that convergence testing without an exact solution only tells us that our discrete solution is converging to some function at the expected rate. In particular, it is possible that our scheme is converging to the *wrong* function at the correct rate. This would not be revealed by the convergence tests described above. Thus, in order to have confidence that our numerical solution is an accurate approximation of the true solution, we must consider other tests of our code. These may include testing the convergence of independent residuals, monitoring constraint violation, and (where possible) comparing to known analytic results.

[7] By our previous argument, we expect that the global error for a kth-order scheme will be $\mathcal{O}(h^k)$.

[8] More generally, we could estimate the global-in-space rate of convergence (i.e. over the entire mesh) by taking an appropriate discrete norm of the various error quantities.

Remark 2.5. Our error analysis relied on the assumption that the error can be expanded as a power series of the form (2.20). In general, one may only be able to prove the existence of such an expansion for very simple PDEs. However, successful convergence tests indicate that this expansion is valid, at least locally. Additionally, the error analysis implicitly assumes that the solution is (suitably) smooth. This assumption does not hold for weak solutions containing discontinuities. In this case the expected order of convergence based on Taylor series expansions will likely not be seen in practice, see for example the discussion in [50, §8.7].

2.3.1. *Exercises.*

(1) Run a convergence test for the upwind method (2.8). Compute the error ratio and plot $\log_2(R)$ over time.

3. Gowdy Cosmologies and the BKL Conjecture

Gowdy spacetimes[9] are the simplest spatially inhomogeneous[10] cosmological models. For this reason, they have been extensively studied over both analytically and numerically, see [1, 2, 11, 14, 17–22, 28, 34, 45, 46, 48, 59, 60, 65] for a small selection of articles. The $\mathbb{T}^3$ Gowdy metric takes the form [11]

$$g = e^{\frac{t-\lambda}{2}}\Big(-e^{2t}dt^2 + dx^2\Big) + e^{-t}\Big(e^P dy^2 + 2e^P Q dy dz + (e^P Q^2 + e^{-P})dz^2\Big), \tag{3.1}$$ eqn:GowdyMetric_A

the functions λ, P, and Q depend only on $(t,x) \in \mathbb{R}\times\mathbb{R}$ and are 2π-periodic in x. In these coordinates, the big bang singularity is approached as $t \to \infty$. Additionally, by setting $Q = 0$ in (3.1), we obtain the subclass of *polarised* Gowdy spacetimes.

Now, by using the Gowdy metric (3.1), the vacuum Einstein equations[11]

$$G_{ab} = 0$$

can be split into three wave equations[12]

$$\partial_t^2 P - e^{-2t}\partial_x^2 P = e^{2P}\big(Q_t^2 - e^{-2t}Q_x^2\big), \tag{3.2}$$ eqn:P_Wave

$$\partial_t^2 Q - e^{-2t}\partial_x^2 Q = 2\Big(e^{-2t}P_x Q_x - P_t Q_t\Big), \tag{3.3}$$ eqn:Q_Wave

$$\partial_t^2 \lambda - e^{-2t}\partial_x^2 \lambda = -2e^{-2t}\Big(P_x^2 + e^{2P}Q_x^2\Big), \tag{3.4}$$ eqn:lambda_Wave

and two constraint equations

$$0 = C_H := \partial_t\lambda - e^{-2t}\Big(P_x^2 + e^{2P}Q_x^2 + e^{2t}P_t^2 + e^{2P+2t}Q_t^2\Big), \tag{3.5}$$ eqn:Hamiltonian_Con

$$0 = C_M := \partial_x\lambda - 2\Big(P_x P_t + e^{2P}Q_x Q_t\Big), \tag{3.6}$$ eqn:Momentum_Constr

where (3.5) and (3.6) are the Hamiltonian and momentum constraints, respectively. Taking time derivatives of (3.5) and (3.6) and using the evolution equations (3.2)-(3.4), a short calculation shows the constraints propagate according to

$$\partial_t C_H - \partial_x(e^{-2t}C_M) = 0, \tag{3.7}$$?eqn:Hamiltonian_Co

$$\partial_t C_M - \partial_x C_H = 0. \tag{3.8}$$?eqn:Momentum_Const

Thus, if the constraints are satisfied initially they will be preserved by the evolution equations (3.2)-(3.4).

[9] See Appendix A for a (brief) discussion of the more general $\mathbb{T}^2$-symmetric spacetimes and the reduction to Gowdy symmetry.

[10] That is, models which vary in space.

[11] Our indexing conventions are as follows: lower case Latin letters, e.g. i, j, k, will label spacetime coordinate indices that run from 0 to 3 while upper case Latin letters, e.g. I, J, K, will label spatial coordinate indices that run from 1 to 3.

[12] These wave equations can be derived by taking appropriate linear combinations of the components of the Einstein tensor G_{ab}.

Note that the wave equations (3.2) and (3.3) decouple from λ. In fact, it is sufficient to solve for these variables alone without initially imposing the constraint (3.5). Recall, however, that the $\mathbb{T}^3$ spatial topology means that the metric variables must be 2π-periodic. By (3.6), this implies that

$$\int_0^{2\pi} 2\Big(P_x P_t + e^{2P} Q_x Q_t \; dx\Big) = 0, \tag{3.9}$$ eqn:integral_condit

which must be imposed on our initial data. Nonetheless, this is a significant simplification from the general Einstein equations. To summarise, the Einstein equations in Gowdy symmetry reduce to the two wave equations (3.2)-(3.3) subject to the condition (3.9) on the initial data.

Since the Einstein equations are significantly simpler in Gowdy symmetry, these spacetimes are a natural place to start when investigating mathematical problems such as the famous *BKL conjecture*. A brief discussion of the BKL conjecture and related work is given below.

3.1. **BKL Conjecture.** Let us consider the dynamical behaviour of cosmological models near the initial 'big bang' singularity (that is, evolving towards the past). Foundational work in this area was conducted by Khalatnikov and Lifshitz (KL) [54], and subsequently extended together with Belinskii (BKL) [7, 8]. In particular, they studied the behaviour of solutions near the initial singularity by neglecting certain terms in the field equations to obtain a more tractable system. Their heuristic results led to the so-called BKL conjecture[13]:

Conjecture 3.1 (BKL)**.** *For a wide class of cosmological solutions of the Einstein equations, a spacelike initial singularity is* ***local****,* ***oscillatory****, and* ***vacuum-dominated****.*

The term local means spatial derivatives become negligible and solutions of the Einstein equations can be well-approximated by spatially homogeneous solutions at each point in space. The singularity is called oscillatory if these (pointwise) spatially homogeneous approximations display highly chaotic, *mixmaster* behaviour similar to the Bianchi VIII and IX models.

By vacuum-dominated, we mean that the (appropriately re-scaled) matter terms in the field equations become negligible near the big bang. This is often expressed by the maxim "matter does not matter". In certain cases, however, the matter content can play a significant role. Indeed, BKL [8] subsequently suggested that a scalar field should suppress oscillations near the big bang. This was further extended by Barrow [6] who argued that stiff fluids ($K = 1$) should have a similar effect, which he referred to as a *quiescent cosmology*. More generally, we can say a singularity is quiescent if the oscillatory behaviour is suppressed [4, 33, 67]. In fact, all rigorous results for the BKL conjecture (outside of spatial homogeneity), to date, are restricted to the quiescent case.

A version of the BKL conjecture was first rigorously established for the vacuum Einstein equations by Isenberg and Moncrief [45] (see also the related work [28]) under a polarised Gowdy symmetry assumption. In this case, they established that the approach to singularity is asymptotically velocity term dominated (AVTD) and non-oscillatory. That is, solutions are well-approximated by the corresponding solutions obtained by setting certain times to zero in field equations. It should be noted that it is the symmetry class, not the presence of matter, which suppresses oscillatory behaviour in this case.

Theorem 3.2 (AVTD for Polarised Gowdy (Rough Version), Isenberg & Moncrief [45])**.** *Every solution of the polarised Gowdy equations ($Q = 0$),*

$$\begin{aligned} \partial_t^2 P - e^{-2t}\partial_x^2 P &= 0, \\ \partial_t \lambda - e^{-2t} P_x^2 - P_t^2 &= 0, \\ \partial_x \lambda - 2P_x P_t &= 0, \end{aligned}$$

[13]See Uggla et al. [71] for a rigorous formulation of the conjecture.

asymptotically approaches a solution of the velocity term dominated (VTD) system

$$\partial_t^2 P = 0,$$
$$\partial_t \lambda - P_t^2 = 0,$$
$$\partial_x \lambda - 2P_x P_t = 0,$$

as $t \to \infty$.

The non-polarised setting was first studied using numerical methods [11, 14, 34, 44]. Here, the spacetimes were observed to monotonically converge towards a Kasner solution, except at isolated points where so-called *spikes* formed. In particular, the existence of spikes indicates that the BKL picture is incomplete. Moreover, it is expected that spikes will also play an important role outside the Gowdy symmetric setting [36, 66]. For further numerical and analytic work on spikes, see [29, 43, 56, 57, 61]. In the following section, we will discuss the numerical simulation of Gowdy spacetimes with spikes.

Subsequent analytic work by Ringström [63–65] has given a detailed description of the asymptotic behaviour in Gowdy spacetimes. In particular, these results demonstrate that $\mathbb{T}^3$ Gowdy spacetimes are, in fact, AVTD.

Significantly less is known about oscillatory behaviour near the big bang outside of spatial homogeneity. The earliest work in this direction is due to Berger, Isenberg, and Weaver who numerically demonstrated that local mixmaster dynamics (LMD) can occur in the inhomogeneous setting for the so-called Magnetic Gowdy spacetimes [76, 77]. Both oscillatory and quiescent behaviour consistent with BKL has since been numerically observed in $\mathbb{T}^2$ and $U(1)$-symmetry [5, 13, 15, 16, 57] as well as in spacetimes without symmetry [30, 35].

As mentioned above, rigorous analytic work on the BKL conjecture is currently limited to quiescent singularities. However, a recent breakthrough in this direction is the result of Fournodavlos, Rodnianski, and Speck [33] which established that the approach to the singularity is AVTD for perturbations of the full range of 'sub-critical' Kasner spacetimes. For subsequent extensions and related works, see [3, 23–25, 27, 32, 41, 52, 53, 67]. Additionally, recent work on weak null singularities [31, 58], which are expected to occur in the interior of black holes, further indicates that BKL conjecture is incomplete. We refer to the reviews [9, 66] for further details on recent progress.

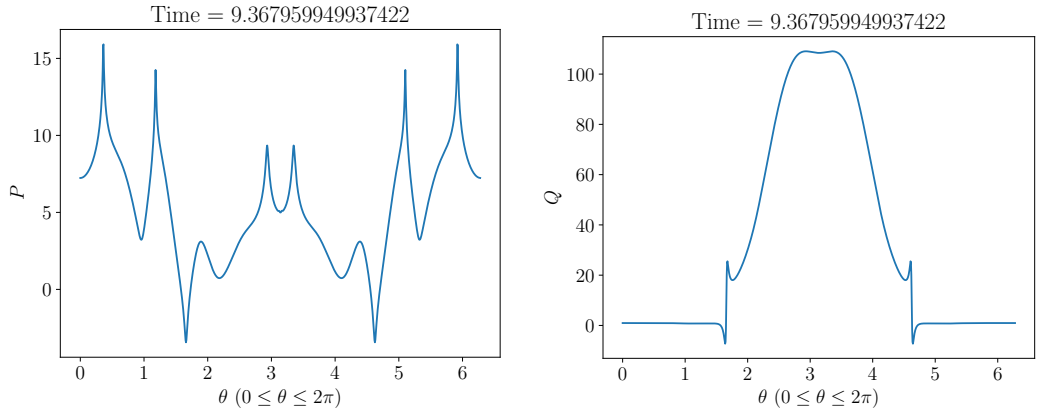
3.2. **Numerical Simulations of Gowdy Spacetimes and Spikes.** The non-polarised Gowdy spacetimes proved significantly more difficult to analytically study than the polarised class. This prompted Berger and Moncrief to form a research program in the early 1990s to numerically investigate the behaviour of cosmological solutions near the big bang, starting with the general $\mathbb{T}^3$ Gowdy class.

Their initial simulations in [14] observed AVTD behaviour towards the singularity except at finitely many points, where small scale features known as spikes developed. Their findings were subsequently replicated and extended in a variety of papers, see for example [11, 36, 56, 57, 61]. In Figure 3(a)-3(b), we show some spikes in Q and P for a numerical solution with the following initial data at $t = 0$,

$$\begin{aligned} P_0 &= 5\cos(x), \\ Q_0 &= 0, \\ P &= 0, \\ Q &= \cos(x). \end{aligned} \tag{3.10}$$?eqn:Example_Gowdy_

What is the cause of these spikes? First, let's consider the AVTD system for P and Q obtained by setting the spatial derivatives to 0 in (3.2)-(3.3),

$$\partial_t^2 P = e^{2P}(\partial_t Q)^2,$$
$$\partial_t^2 Q = -2\partial_t P \partial_t Q.$$



(a) P (b) Q

FIGURE 3. Spiky features in P and Q at $t \approx 9.36$. $N = 1000$.

Expressed in first order form, this becomes

$$\partial_t P = P_0, \tag{3.11}$$ eqn:P_AVTD

$$\partial_t P_0 = e^{2P} Q_0^2, \tag{3.12}$$ eqn:P0_AVTD

$$\partial_t Q = Q_0, \tag{3.13}$$ eqn:Q_AVTD

$$\partial_t Q_0 = -2 Q_0 P_0. \tag{3.14}$$ eqn:Q0_AVTD

Using the AVTD equations (3.11)-(3.14), we can make a heuristic argument for why spiky features develop in Q and P.

First, let's consider spikes in Q, such as the feature around $x = 1.6$ in Figure 3(b). There are three cases to consider

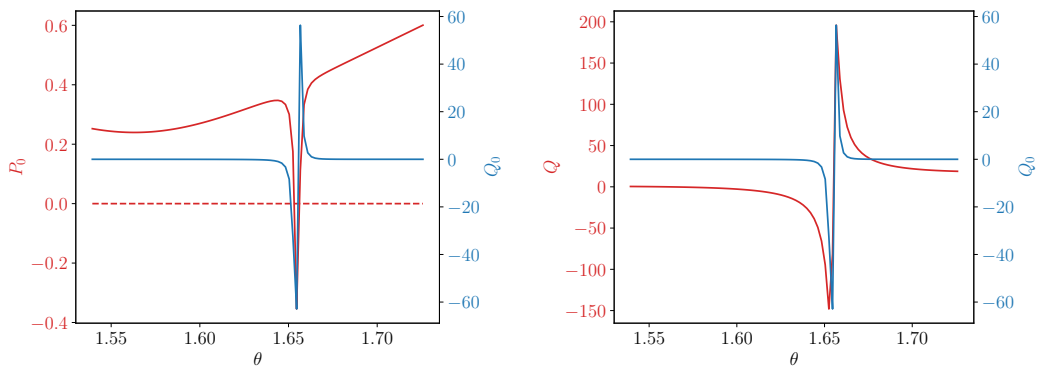
(1) $Q_0 = 0$,
(2) $P_0 < 0$, $Q_0 < 0$,
(3) $P_0 < 0$, $Q_0 > 0$.

Case 1. By (3.13), Q will remain constant when $Q_0 = 0$.

Case 2. If P_0 and Q_0 are negative, then the right hand side of (3.14) will be negative, so Q_0 will decrease. Since $Q_0 < 0$, (3.13) also implies that $Q \searrow -\infty$. However, since the right hand side of (3.12) is always positive, P_0 is growing and at some point the quantity $-2Q_0P_0$ will become positive, causing both Q_0 to grow towards (but not cross) zero.

Case 3. If $P_0 < 0$ and $Q_0 > 0$, then the right hand side of (3.14) will be positive, so Q_0 will increase. Since $Q_0 > 0$ and increasing, $Q \nearrow \infty$. As in case 2, P_0 is increasing and so $-2Q_0P_0$ will eventually switch sign. This will cause Q_0 to decay towards zero.

How would these three cases combine to cause a spike in Q? Suppose, we had two regions on the domain. One where $Q_0 < 0$ $P_0 < 0$ and one where $Q_0 > 0$ and $P_0 < 0$. By continuity, there must be a point between the two regions where $Q_0 = 0$. At this point, by case 1, the value of Q will be fixed. On either side, the value of Q will approach $\pm\infty$. The transition between these regions will cause a discontinuous, spiky looking feature in Q. Indeed, our numerical results are consistent with this heuristic argument, see for example Figures 4(a)-4(b).



(a) P_0 (red) and Q_0 (blue). The red dashed line marks $P_0 = 0$.

(b) Q (red) and Q_0 (blue).

FIGURE 4. The values of Q_0 and P_0 around a spike in Q at $t \approx 15.75$. $N = 3000$. Observe that around the spike, $P_0 < 0$ and Q_0 transitions from negative to positive.

Now, let's turn to the downward spikes in P. As argued above, P_0 will eventually become positive, aside from points where Q_0 vanishes. This, in turn, will drive $P \nearrow \infty$. However, at the special point where Q_0 vanishes, P_0 and P can *remain* negative. The transition between these behaviours would lead to a downward pointing spike in P.

In both of the above cases we expect that spatial derivatives would eventually become large around the spike points, at which point the AVTD approximation may no longer be valid, see for example the discussion in [55, §7.2].

It remains to discuss the upward pointing spikes in P. Suppose that $P > 0$ is large and $P_0 > 1$. In this case, we expect that the evolution for P_0 is approximated by

$$\partial_t P_0 = -e^{2(P-t)} Q_x^2. \tag{3.15}$$

eqn:P_Upward_Spike_

This is because the exponential term $e^{2(P-t)}$ is no longer negligible and, as discussed above, we expect that Q_0 will be driven towards zero when $P_0 > 0$. Since the right hand side of (3.15) is negative, P_0 should eventually be driven below 1. This would lead to $e^{2(P-t)} Q_x^2$ becoming negligible and the growth rate of P slowing. However, at special points where $Q_x = 0$, there is no mechanism to drive $P_0 < 1$. Hence, at this point P will grow faster than adjacent points in space resulting in an upward pointing spike.

It is important to emphasise that the above arguments are, of course, heuristic and require significantly more work to make them rigorous, cf. [65].

Appendix A. Notes on $\mathbb{T}^2$-Symmetric Spacetimes

⟨app:T2_Spacetimes⟩ In the following, we introduce general $\mathbb{T}^2$-symmetric spacetimes and discuss the reduction to Gowdy symmetry.

A.1. G_2 **Spacetimes.** Cosmological models which admit a two dimensional isometry group that acts simply transitively on spacelike hypersurfaces are known as G_2 models. In this case, the field equations reduce to a system of PDEs which depend on time and one spatial variable. Thus, G_2 models are the simplest example of spatially inhomogeneous cosmologies.

Throughout the remainder of this section, we will focus on the case where a $\mathbb{T}^2$-isometry group acts on spacelike hypersurfaces with $\mathbb{T}^3$-topology. In particular, this means the field equations reduce to a $1+1$ system with periodic boundary conditions. Using coordinates adapted to the Killing vectors[14], a general $\mathbb{T}^2$-symmetric spacetime can be written in *areal* coordinates[15] as (cf. [10, 13])

$$g = e^{2(\eta-U)}(-e^{2\alpha}dt^2 + d\theta^2) + e^{2U}(dx + Ady + (\tilde{G} + A\tilde{H})d\theta)^2 + e^{-2(U+t)}(dy + \tilde{H}d\theta)^2, \tag{A.1}$$ eqn:T2_SymmetricMet

where the metric functions η, U, α, A, $\tilde{G}$, and $\tilde{H}$ depend only on $(t,\theta) \in \mathbb{R}\times\mathbb{R}$ and are 2π-periodic in θ. Our coordinates are called areal, as the time coordinate t is proportional to the area of the symmetry orbits. That is $t \simeq \sqrt{\gamma}$ where γ is the Riemannian two-metric

$$\gamma = e^{2U}dx^2 + 2e^{2U}Adxdy + (e^{2U}A^2 + e^{-2U-2t})dy^2. \tag{A.2}$$ {?}

For a $\mathbb{T}^2$-symmetric spacetime, we can also define the 'twists'

$$J_{(X)} = \epsilon_{\mu\nu\gamma\delta}X^\mu Y^\nu \nabla^\gamma X^\delta = -e^{4U-2\eta-\alpha-t}(\tilde{G}_t + A\tilde{H}_t), \tag{A.3}$$ eqn:J_X_definition

$$\begin{aligned} J_{(Y)} = \epsilon_{\mu\nu\gamma\delta}X^\mu Y^\nu \nabla^\gamma Y^\delta &= -e^{-3t-2\eta-\alpha}\big(\tilde{H}_t + e^{2t+4U}A(\tilde{G}_t + A\tilde{H}_t)\big) \\ &= -e^{-3t-2\eta-\alpha}\tilde{H}_t + AJ_{(X)}, \end{aligned} \tag{A.4}$$ eqn:J_Y_definition

where the subscript t denotes differentiation with respect to the time coordinate, $\epsilon_{\mu\nu\gamma\delta}$ is the volume form, and $X = \partial_x$ and $Y = \partial_y$ are the Killing vectors of the spacetime. The twists $J_{(X)}$ and $J_{(Y)}$ are often called the 'twist constants', which comes from the fact that these quantities are constant for the vacuum Einstein equations. When coupling to matter, however, the twists are not necessarily constant.

A spacetime is Gowdy symmetric [39, 40] if both of the twists vanish. In this case, the group action is said to be *orthogonally transitive* which means the two-dimensional subspaces orthogonal to group orbits are surface-forming [37, 74]. To see that this is equivalent to $J_{(X)}$ and $J_{(Y)}$ vanishing, let us consider the two dimensional subspace V_x^* of the dual space of T_x^*M generated by the Killing vector fields[16] X, Y at a point x. This implicitly defines the two dimensional subspace of orthogonal vectors $W_x \subset T_xM$,

$$v^\mu X_\mu = 0, \quad v^\mu \in W_x. \tag{A.5}$$ {?}

Next, we denote the associated bundles by W and V^*. We want to know when W is surface forming. That is, when does W admit an integral submanifold[17]? By Frobenius' theorem [75, §B.3], W is surface-forming if and only if for all $\omega_\alpha \in V^*$ there exist one-forms $v_\alpha^{(a)}$ such that

$$\mathrm{d}\omega = \sum_{a=1}^{2} v^{(a)} \wedge \mu^{(a)} \tag{A.6}$$ eqn:Frobenius_Dual_

[14] Here, we mean that $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$ are (spacelike) Killing vectors.

[15] For previous work using areal coordinates, see [10–12, 14, 48].

[16] That is, generated by their associated one-forms defined by lowering indices using the metric.

[17] An integral submanifold is an embedded submanifold S such that at each point $y \in S$ the tangent space coincides with W_y.

where each $\mu^{(a)} \in V^*$. Now, since V^* is spanned by X and Y, we can replace the $\mu^{(a)}$ with a linear combination of X and Y. Thus, (A.6) is equivalent to

$$\mathrm{d}X = X \wedge \tilde{v}^{(1)} + Y \wedge \tilde{v}^{(2)}, \tag{A.7}$$ eqn:X_frob_dual

$$\mathrm{d}Y = X \wedge \bar{v}^{(1)} + Y \wedge \bar{v}^{(2)}, \tag{A.8}$$ eqn:Y_frob_dual

where $\tilde{v}^{(a)}$ and $\bar{v}^{(a)}$ are 'arbitrary' one-forms. Taking the wedge products of (A.7)-(A.8) with X and Y yields[18]

$$X \wedge Y \wedge \mathrm{d}X = 0, \tag{A.9}$$ eqn:X_frob_dual2

$$X \wedge Y \wedge \mathrm{d}Y = 0. \tag{A.10}$$ eqn:Y_frob_dual2

A short calculation[19] then shows that (A.9)-(A.10) is equivalent to

$$X_{[\alpha} Y_{\beta} \nabla_{\gamma} X_{\delta]} = 0,$$
$$X_{[\alpha} Y_{\beta} \nabla_{\gamma} Y_{\delta]} = 0,$$

which, in turn, is clearly equivalent to the vanishing of the twists.

A.2. **Einstein Equations.** Let us now derive the Einstein equations for the metric (A.1). First, using (A.3) and (A.4), we obtain the following evolution equations for $\tilde{G}$ and $\tilde{H}$

$$\partial_t \tilde{H} = e^{3t+2\eta+\alpha}\big(AJ_{(X)} - J_{(Y)}\big), \tag{A.11}$$ eqn:H_evoT2

$$\partial_t \tilde{G} = -e^{t+\alpha+2\eta}\Big(e^{-4U} J_{(X)} - e^{2t} A\big(AJ_{(X)} - J_{(Y)}\big)\Big). \tag{A.12}$$ eqn:G_evoT2

Next, let us consider the Einstein equations for an arbitrary matter source

$$G_{\mu\nu} = T_{\mu\nu}. \tag{A.13}$$ {?}

We find, after a straightforward calculation, that the Einstein equations in $\mathbb{T}^2$-symmetry consist of three wave equations

$$\begin{aligned}\partial_{tt} A &= e^{2\alpha}\partial_{\theta\theta} A + e^{2\alpha}\partial_\theta A \partial_\theta \alpha - \partial_t A + 4e^{2\alpha}\partial_\theta A \partial_\theta U - 4\partial_t A \partial_t U + \partial_t A \partial_t \alpha \\ &\quad + 2e^{-4U+2(\alpha+\eta)}(T_{23} - AT_{22}) - e^{2(t+\eta+\alpha)-4U} J_{(X)}\big(AJ_{(X)} - J_{(Y)}\big),\end{aligned} \tag{A.14}$$ eqn:A_wave_eqnT2

$$\begin{aligned}\partial_{tt} U &= \frac{1}{2} + e^{2\alpha}\partial_\theta U \partial_\theta \alpha + e^{2\alpha}\partial_{\theta\theta} U + \frac{1}{2} e^{2t+4U}\Big((\partial_t A)^2 - e^{2\alpha}(\partial_\theta A)^2\Big) + \partial_t U\big(1 + \partial_t \alpha\big) \\ &\quad + \frac{1}{2}\partial_t \alpha + \frac{1}{2} e^{-4U+2(\alpha+\eta)} T_{22} - \frac{1}{2} e^{2(t+\alpha+\eta)} A^2 T_{22} + e^{2(t+\alpha+\eta)} AT_{23} - \frac{1}{2} e^{2t+2\alpha+2\eta} T_{33} \\ &\quad + \frac{1}{4} e^{-4U+2(\eta+\alpha+t)} J_{(X)}^2 - \frac{1}{4} e^{4t+2(\eta+\alpha)}\big(-AJ_{(X)} + J_{(Y)}\big)^2,\end{aligned} \tag{A.15}$$?eqn:U_wave_eqnT2?

$$\begin{aligned}\partial_{tt}\eta &= -e^{2(t+\alpha+\eta)} A^2 T_{22} + 2e^{2(t+\alpha+\eta)} AT_{23} - e^{2(t+\alpha+\eta)} T_{33} - \frac{1}{4} e^{4U+2(t+\alpha)}(\partial_\theta A)^2 \\ &\quad + e^{2\alpha}(\partial_\theta U)^2 + e^{2\alpha}(\partial_\theta \alpha)^2 + e^{2\alpha}(\partial_\theta \alpha \partial_\theta \eta) + e^{2\alpha}\partial_{\theta\theta}\alpha + e^{2\alpha}\partial_{\theta\theta}\eta + \frac{1}{4} e^{4U+2t}(\partial_t A)^2 \\ &\quad - (\partial_t U)^2 + \partial_t \alpha \partial_t \eta - \frac{1}{4} e^{-4U+2(\eta+\alpha+t)} J_{(X)}^2 - \frac{3}{4} e^{4t+2(\eta+\alpha)}\big(-AJ_{(X)} + J_{(Y)}\big)^2,\end{aligned} \tag{A.16}$$?eqn:eta_wave_eqnT2

and seven first order equations

$$\begin{aligned}\partial_t \eta &= -\frac{1}{4} e^{4U+2t}\Big((\partial_t A)^2 + e^{2\alpha}(\partial_\theta A)^2\Big) - \Big((\partial_t U)^2 + e^{2\alpha}(\partial_\theta U)^2\Big) - T_{00}, \\ &\quad - \frac{1}{4} e^{-4U+2(\eta+\alpha+t)} J_{(X)}^2 - \frac{1}{4} e^{4t+2(\eta+\alpha)}\big(-AJ_{(X)} + J_{(Y)}\big)^2,\end{aligned} \tag{A.17}$$ eqn:Hamiltonian_Cor

$$\begin{aligned}\partial_\theta \eta &= -\partial_\theta \alpha + \frac{1}{2} e^{t+\alpha}\tilde{H}\big(A\partial_\theta J_{(X)} - \partial_\theta J_{(Y)}\big) - \frac{1}{2} e^{2t+4U}\partial_\theta A \partial_t A - 2\partial_\theta U \partial_t U \\ &\quad - \frac{1}{2} e^{t+\alpha}\partial_\theta J_{(X)} - T_{01},\end{aligned} \tag{A.18}$$ eqn:momentum_constr

[18] Here, we have used the fact that $A \wedge A = 0$ and $A \wedge B = -B \wedge A$.

[19] Since the exterior derivative is independent of the choice of derivative operator, we can compute it using ∇.

$$\partial_t \alpha = -1 + T_{00} - e^{2\alpha} T_{11} + 2e^{2\alpha}\tilde{G}T_{12} + 2e^{2\alpha}\tilde{H}T_{13} - e^{2\alpha}\tilde{G}^2 T_{22} - 2e^{2\alpha}\tilde{G}\tilde{H}T_{23}$$
$$- e^{2\alpha}\tilde{H}^2 T_{33} + \frac{1}{2}e^{-4U+2(\eta+\alpha+t)} J_{(X)}^2 + \frac{1}{2}e^{4t+2(\eta+\alpha)}(-AJ_{(X)} + J_{(Y)})^2, \tag{A.19}$$?eqn:alpha_evo_T2?

$$\partial_\theta J_{(X)} = 2e^{-(t+\alpha)} T_{02}, \tag{A.20}$$ eqn:J_X_constraint

$$\partial_\theta J_{(Y)} = 2e^{-(t+\alpha)} T_{03}, \tag{A.21}$$ eqn:J_Y_constraint

$$\partial_t J_{(X)} = 2e^{\alpha - t}(\tilde{H}T_{23} + \tilde{G}T_{22} - T_{12}), \tag{A.22}$$ eqn:J_X_evo

$$\partial_t J_{(Y)} = 2e^{\alpha - t}(\tilde{H}T_{33} + \tilde{G}T_{23} - T_{13}), \tag{A.23}$$ eqn:J_Y_evo

where (A.17) is the Hamiltonian constraint and (A.18), (A.20), and (A.21) are the components of the momentum constraint. Hence, the full system of equations is given by (A.11)-(A.12) and (A.14)-(A.22).

It is clear that (A.20)-(A.23) imply the twist quantities $J_{(X)}$ and $J_{(Y)}$ are constant when $T_{\mu\nu} = 0$. In particular, if they are initially zero, then they must remain zero. From (A.11)-(A.12), we then find that $\tilde{H}$ and $\tilde{G}$ must also be constant.

If matter is present, Gowdy symmetry additionally requires

$$T_{02} = T_{03} = (\tilde{H}T_{23} + \tilde{G}T_{22} - T_{12}) = (\tilde{H}T_{33} + \tilde{G}T_{23} - T_{13}) = 0. \tag{A.24}$$ {?}

For a perfect fluid, these conditions imply the fluid four-velocity u only has two non-zero components

$$u = u_0 dt + u_1 d\theta. \tag{A.25}$$ {?}

A.3. **Relation to the Orthonormal Frame Formalism.** To conclude this section, we explicitly give the relation between the orthonormal frame formalism, see for example [72], and the metric formulation derived above. From inspection of (A.1), we see that an orthogonal dual frame θ^i is given by

$$\theta^0 = e^{\eta+\alpha-U}dt, \;\; \theta^1 = e^{\eta-U}d\theta,$$
$$\theta^2 = e^U(dx + Ady + (\tilde{G} + A\tilde{H})d\theta), \;\; \theta^3 = e^{-(U+t)}(dy + \tilde{H}d\theta). \tag{A.26}$$ {?}

Recall that our frame fields must satisfy the conditions[20]

$$\theta^i(e_j) = \delta^i_j, \quad \theta^0(e_0) = -e_0^\flat(e_0) = 1,$$
$$\theta^A(e_A) = e_A^\flat(e_A) = 1, \quad \theta^0_\mu = -e_{0\mu}, \quad \theta^A_\alpha = e_{A\alpha}. \tag{A.27}$$ {?}

By combining these identities with the expressions for the dual frames, we see that the corresponding orthonormal frame vectors are given by

$$e_0 = e^{U-\eta-\alpha}\partial_t, \;\; e_1 = e^{U-\eta}(\partial_\theta - \tilde{G}\partial_x - \tilde{H}\partial_y)$$
$$e_2 = e^{-U}\partial_x, \;\; e_3 = -e^{t+U}A\partial_x + e^{t+U}\partial_y.$$

Next, using the decomposition of the commutator coefficients [68, Page 6]

$$H = -\frac{1}{3}c_0{}^A{}_A, \;\; \sigma_{AB} = -c_0{}^C{}_{<A}\delta_{B>C}, \;\; \dot{u}_A = c_0{}^0{}_A,$$
$$w_A = \frac{1}{4}\epsilon_A{}^{BC}c_B{}^0{}_C, \;\; w_A + \Omega_A = \frac{1}{2}\epsilon_{AB}{}^C c_0{}^B{}_C, \;\; n^{AB} = \frac{1}{2}\epsilon^{ab(A}c_a{}^{B)}{}_b, \;\; a_A = \frac{1}{2}c_A{}^B{}_B,$$

the following identity for the commutator coefficients

$$c_a{}^c{}_b = \omega_a{}^c{}_b - \omega_b{}^c{}_a,$$

and the formula for the connection coefficients [75, Page 50],

$$\omega_{abc} = e^\gamma_a e^\mu_b \nabla_\gamma e_{c\mu},$$

[20]Here $\flat$ denotes the musical isomorphism, see for example [38, §2.3.3].

we obtain the identities

$$
\begin{aligned}
a_1 = a_2 = a_3 = n_{11} = n_{12} = n_{13} = n_{33} = w_1 = w_2 = w_3 = 0,\\
\dot{u}_1 = e^{U-\eta}(-U_\theta + \alpha_\theta + \eta_\theta), \quad H = \frac{1}{3}e^{U-\alpha-\eta}(-1 - U_t + \eta_t),\\
\sigma_{11} = \frac{1}{3}e^{U-\alpha-\eta}(1 - 2U_t + 2\eta_t), \quad \sigma_{12} = \frac{1}{2}e^{3U-\alpha-2\eta}(\tilde{G}_t + A\tilde{H}_t),\\
\sigma_{13} = \frac{1}{2}e^{-t+U-\alpha-2\eta}\tilde{H}_t, \quad \sigma_{22} = \frac{1}{3}e^{U-\alpha-\eta}(1 + 4U_t - \eta_t),\\
\sigma_{23} = \frac{1}{2}e^{t+3U-\alpha-\eta}A_t, \quad \sigma_{33} = -\frac{1}{3}e^{U-\alpha-\eta}(2 + 2U_t + \eta_t),\\
n_{22} = e^{t+3U-\eta}A_\theta, \quad n_{23} = -e^{U-\eta}U_\theta,\\
\Omega_1 = \frac{-1}{2}e^{t+3U-\alpha-\eta}A_t, \quad \Omega_2 = -\frac{1}{2}e^{-t+U-\alpha-2\eta}\tilde{H}_t, \quad \Omega_3 = \frac{1}{2}e^{3U-\alpha-2\eta}(\tilde{G}_t + A\tilde{H}_t).
\end{aligned}
\tag{A.28}
$$

?eqn:T2_FrameDecomp

The spatial gauge is determined through the conditions

$$
n_{33} = 0, \;\; \Omega_1 = -\sigma_{23}. \tag{A.29}
$$

{?}

In particular, this is the same spatial gauge condition given in equation (68) of [73]. Similarly, the condition $a_1 = 0$ means we are working in the *timelike* areal gauge, see [73] for details.

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Institute of Applied and Computational Mathematics, FORTH, 70013 Heraklion, Greece
Email address: `elliot.marshall@iacm.forth.gr`